

# ADITYA GOSWAMI

Full-Stack Web Developer

## CONTACT

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Full-Stack Web Developer building web applications with React.js, JavaScript, Node.js, Express.js, HTML5 and Tailwind CSS, with MongoDB and MySQL databases. Built StudyRewind, a full-stack application with a FastAPI and PostgreSQL backend, JWT authentication and RAG-style semantic search. Additional practical knowledge of prompt engineering, Retrieval-Augmented Generation (RAG), vector databases, embeddings and AI-assisted development.

## SKILLS

**Programming Languages:** JavaScript, C++, Python, C

**Frontend Development:** React.js, HTML5, Tailwind CSS

**Backend Development:** Node.js, Express.js, FastAPI, JWT Authentication, SQLAlchemy 2.0, Alembic

**Databases:** MongoDB, MySQL, PostgreSQL, pgvector

**AI / LLM:** Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Databases, Embeddings, Semantic Search, Faster-Whisper, Sentence Transformers

**AI-Assisted Development:** Strong working knowledge of AI tools such as Claude, Codex, GitHub Copilot and Antigravity, with the ability to use them efficiently and effectively across the development workflow

**Computer Science:** Data Structures and Algorithms, Object-Oriented Programming (OOP), Computer Networks, DBMS, Software Engineering

## EDUCATION

**Bachelor of Technology, Computer Science and Business System** | 2024 - 2028

Techno International New Town, Kolkata, India (Affiliated with MAKAUT - Maulana Abul Kalam Azad University of Technology)  
Currently in 3rd Year, 5th Semester

**Class XII** | 2024

Sri Chaitanya Techno School

## PROJECTS

### StudyRewind — AI-Powered Educational Search Assistant (Final Year Project)

Built a full-stack web application that ingests YouTube lecture playlists and PDF study materials, transcribes and embeds content locally using AI, and provides semantic search with timestamp and page provenance.

**Backend:** FastAPI (Python), PostgreSQL, pgvector, SQLAlchemy 2.0, Alembic, Argon2id authentication, JWT authorization

**AI/ML:** faster-whisper (local speech transcription), sentence-transformers (paraphrase-multilingual-MiniLM-L12-v2, 384-d embeddings), pgvector cosine similarity search, yt-dlp, youtube-transcript-api, pypdf

**Frontend:** React 18, TypeScript, Vite

#### Key engineering work:

- Implemented a three-step transcript extraction pipeline: YouTube captions → yt-dlp subtitle download → local Whisper AI fallback
- Built a local embedding pipeline generating 384-dimensional multilingual vectors (Hindi + English) with batch processing, idempotency, and validation
- Implemented strict user data isolation with IDOR prevention and UUID-based ownership chains across all API endpoints
- Wrote 184 passing automated tests covering authentication, subject management, transcription, PDF processing, semantic search, and security boundaries
- Resolved a Windows file-locking bug in temporary audio cleanup using bounded retry with read-only attribute handling

### Portfolio — macOS-Inspired Interactive Developer Portfolio

HTML5 • CSS3 • JavaScript • LLM/AI Integration (Planned)

Built an interactive macOS-inspired personal portfolio that presents my profile, projects, skills, experience, education, certificates, resume, and contact information through a desktop-style interface. Implemented a responsive desktop environment with custom wallpaper, menu bar, desktop folders, application Dock, draggable portfolio windows, window controls, window stacking, and responsive layouts using HTML, CSS, and JavaScript.

**Future AI/LLM Integration:** Planning to build a personal LLM-powered portfolio assistant that will be grounded in my portfolio data, allowing recruiters and visitors to ask questions about my skills, projects, experience, education, and technical background instead of manually searching through the portfolio. I also plan to add a job-description matching feature, where hiring managers can paste a job description and the system will use an LLM to compare the requirements against my portfolio projects and skills, providing a structured match breakdown and score. LLM engineering, model integration, and the underlying implementation are currently part of my learning roadmap and are planned for a future version.